A Stam-type inequality for a finite-free convolution and the functional Φ_n

Abstract

We consider the binary operation $\boxplus_n$ on monic degree-n polynomials defined by an explicit coefficient formula (equivalently, by a constant-coefficient differential operator). For a monic polynomial $p(x) = \prod_{i=1}^{n}(x - \lambda_i)$ with simple roots we study

$$\Phi_n(p) \coloneqq \sum_{i=1}^{n} \left(\sum_{j \neq i} \frac{1}{\lambda_i - \lambda_j} \right)^2, \qquad \Phi_n(p) \coloneqq \infty \text{ if } p \text{ has a multiple root.}$$

The question asks whether, for real-rooted p, q,

$$\frac{1}{\Phi_n(p \boxplus_n q)} \;\geq\; \frac{1}{\Phi_n(p)} + \frac{1}{\Phi_n(q)}.$$

We explain structural identities for Φ_n and for $\boxplus_n$, and we present a proof strategy in the spirit of Stam's classical inequality. The key analytic step is a finite-free "score" identity; under this identity, the desired inequality follows by a sharp Cauchy–Schwarz minimization exactly as in the classical Blachman–Stam argument.

1 The operation $\boxplus_n$

Let $p(x) = \sum_{k=0}^{n} a_k x^{n-k}$ and $q(x) = \sum_{k=0}^{n} b_k x^{n-k}$ be monic (so $a_0 = b_0 = 1$). Define

$$(p \boxplus_n q)(x) = \sum_{k=0}^{n} c_k x^{n-k}, \qquad c_k = \sum_{i+j=k} \frac{(n-i)!(n-j)!}{n!(n-k)!}\, a_i b_j.$$

It is convenient to record an equivalent differential-operator form.

Proposition 1 (Differential-operator representation)**.** *Define the constant-coefficient operator*

$$\mathcal{D}_q \coloneqq \sum_{k=0}^{n} \frac{(n-k)!}{n!}\, b_k \frac{d^k}{dx^k}.$$

Then for every monic degree-n polynomial p,

$$p \boxplus_n q \;=\; \mathcal{D}_q p \;=\; \sum_{k=0}^{n} \frac{q^{(n-k)}(0)}{n!}\, p^{(k)}. \tag{1}$$

Proof. Since $q^{(n-k)}(0) = (n-k)!\, b_k$, the second equality in (**??**) is just a rewrite of $\mathcal{D}_q$. Write $p^{(k)}(x) = \sum_{r=0}^{n-k} a_r (n-r)\cdots(n-r-k+1)\, x^{n-r-k}$. The coefficient of x^{n-k} in $\mathcal{D}_q p$ is

$$\sum_{i+j=k} \frac{(n-j)!}{n!}\, b_j \cdot a_i \cdot \frac{(n-i)!}{(n-k)!},$$

which is precisely c_k. $\square$

A useful algebraic way to see commutativity/associativity is via the truncated power-series symbol. Set

$$\mathcal{E}_n(p)(z) := \sum_{k=0}^{n} a_k \frac{(n-k)!}{n!} z^k \in \mathbb{R}[z]/(z^{n+1}).$$

Then one checks directly from the definition of c_k that

$$\mathcal{E}_n(p \boxplus_n q) \equiv \mathcal{E}_n(p)\, \mathcal{E}_n(q) \pmod{z^{n+1}}. \tag{2}$$

In particular, $\boxplus_n$ is commutative and associative.

2 The functional Φ_n

Let $p(x) = \prod_{i=1}^{n}(x - \lambda_i)$ be monic. When the roots are simple we define

$$\Phi_n(p) = \sum_{i=1}^{n} \left(\sum_{j \neq i} \frac{1}{\lambda_i - \lambda_j} \right)^2.$$

Lemma 1 (Two equivalent formulas)**.** *Assume p has simple roots $\lambda_1, \ldots, \lambda_n$ (not necessarily real). Then*

$$\Phi_n(p) = \sum_{i \neq j} \frac{1}{(\lambda_i - \lambda_j)^2}, \tag{3}$$

$$\Phi_n(p) = \frac{1}{4} \sum_{i=1}^{n} \left(\frac{p''(\lambda_i)}{p'(\lambda_i)} \right)^2. \tag{4}$$

Proof. For (**??**), expand

$$\sum_{i=1}^{n} \left(\sum_{j \neq i} \frac{1}{\lambda_i - \lambda_j} \right)^2 = \sum_{i=1}^{n} \sum_{j \neq i} \frac{1}{(\lambda_i - \lambda_j)^2} + \sum_{i=1}^{n} \sum_{\substack{j,k \neq i \\ j \neq k}} \frac{1}{\lambda_i - \lambda_j} \cdot \frac{1}{\lambda_i - \lambda_k}.$$

Fix distinct i, j, k. The contribution of the triple (i, j, k) to the second sum plus the corresponding contributions for cyclic permutations equals

$$\frac{1}{\lambda_i - \lambda_j} \frac{1}{\lambda_i - \lambda_k} + \frac{1}{\lambda_j - \lambda_i} \frac{1}{\lambda_j - \lambda_k} + \frac{1}{\lambda_k - \lambda_i} \frac{1}{\lambda_k - \lambda_j} = 0,$$

which is a direct algebraic identity (bring to a common denominator and note the numerator cancels). Thus all cross terms cancel and (**??**) follows.

For (**??**), write $p(x) = (x - \lambda_i) r_i(x)$ with $r_i(\lambda_i) = p'(\lambda_i)$. Then $p'(x) = r_i(x) + (x - \lambda_i) r_i'(x)$ and $p''(x) = 2r_i'(x) + (x - \lambda_i) r_i''(x)$, hence

$$\frac{p''(\lambda_i)}{p'(\lambda_i)} = 2 \frac{r_i'(\lambda_i)}{r_i(\lambda_i)} = 2 \sum_{j \neq i} \frac{1}{\lambda_i - \lambda_j}.$$

Squaring and summing over i yields (**??**). □

Corollary 1 (Positivity and scaling)**.** *If p has simple real roots, then $\Phi_n(p) > 0$. If $p_s(x) := s^{-n} p(sx)$ (so the roots scale by $1/s$), then*

$$\Phi_n(p_s) = s^2\, \Phi_n(p), \qquad \frac{1}{\Phi_n(p_s)} = \frac{1}{s^2} \frac{1}{\Phi_n(p)}.$$

3 A matrix model for $\boxplus_n$

Although the inequality in the question is purely algebraic, it is helpful to note a standard matrix interpretation of $\boxplus_n$ (which also explains real-root preservation).

Proposition 2 (Expected characteristic polynomial model)**.** *Let $A = \mathrm{diag}(\lambda_1, \dots, \lambda_n)$ and $B = \mathrm{diag}(\mu_1, \dots, \mu_n)$ be real diagonal matrices and set*

$$p(x) = \det(xI - A), \qquad q(x) = \det(xI - B).$$

Let U be Haar-uniform on the orthogonal group $O(n)$ and put $M \coloneqq A + UBU^\top$. Then

$$\mathbb{E}\left[\det(xI - M)\right] \;=\; p \boxplus_n q(x).$$

Proof. Write the characteristic polynomial as

$$\det(xI - M) = \sum_{k=0}^{n} (-1)^k e_k(M)\, x^{n-k},$$

where $e_k(M)$ is the sum of all $k \times k$ principal minors of M. Fix a subset $S \subseteq \{1, \dots, n\}$ with $|S| = k$; write M_S for the principal $k \times k$ submatrix indexed by S. Since A_S is diagonal, the determinant expansion along fixed points gives the exact identity

$$\det(M_S) = \sum_{T \subseteq S} \left(\prod_{i \in T} \lambda_i \right) \det\left((UBU^\top)_{S \setminus T} \right). \tag{5}$$

Taking expectations and using Haar invariance under permutation matrices, $\mathbb{E}[\det((UBU^\top)_R)]$ depends only on $|R| = j$. Denote this common value by $\alpha_{n,j}(B)$. By orthogonal invariance, $\alpha_{n,j}(B)$ must be a symmetric polynomial in the eigenvalues of B of degree j, hence it is proportional to $e_j(B) = e_j(\mu_1, \dots, \mu_n)$. Evaluating at $B = I$ gives $\alpha_{n,j}(I) = 1$ while $e_j(I) = \binom{n}{j}$, so

$$\mathbb{E}[\det((UBU^\top)_R)] = \frac{e_j(B)}{\binom{n}{j}}.$$

Now sum (**??**) over all S of size k, group terms by $|T| = i$ and $|S \setminus T| = j$ (so $i + j = k$), and note that

$$\sum_{\substack{T \subseteq \{1,\dots,n\} \\ |T| = i}} \prod_{t \in T} \lambda_t = e_i(A), \qquad \#\{R \subseteq \{1, \dots, n\} \setminus T : |R| = j\} = \binom{n-i}{j}.$$

Thus

$$\mathbb{E}[e_k(M)] = \sum_{i+j=k} e_i(A) \binom{n-i}{j} \frac{e_j(B)}{\binom{n}{j}}.$$

Using $e_i(A) = (-1)^i a_i$ and $e_j(B) = (-1)^j b_j$ and multiplying by $(-1)^k$ to obtain the coefficient of x^{n-k}, we conclude

$$c_k = \sum_{i+j=k} a_i b_j \frac{\binom{n-i}{j}}{\binom{n}{j}} = \sum_{i+j=k} a_i b_j \frac{(n-i)!(n-j)!}{n!(n-k)!},$$

which is exactly the definition of $\boxplus_n$. $\square$

4 The Stam-type inequality

We now return to the question:

Theorem 1 (Finite-free Stam inequality)**.** *Let p, q be monic, real-rooted polynomials of degree n. Then*

$$\frac{1}{\Phi_n(p \boxplus_n q)} \geq \frac{1}{\Phi_n(p)} + \frac{1}{\Phi_n(q)},$$

where $1/\infty$ is interpreted as 0.

Comments on the proof

The inequality in Theorem **??** is a finite-n analogue of Stam's inequality for Fisher information. A standard proof of the classical Stam inequality is due to Blachman and Carlen and proceeds from two inputs:

1. a *score identity* stating that the score of a sum is the conditional expectation of each summand's score given the sum;
2. the Cauchy–Schwarz inequality plus a one-parameter minimization.

The finite-free setting is more algebraic: the "law" is encoded by the expected characteristic polynomial, and Φ_n is an intrinsic quantity associated with its roots (Lemma **??**). A finite-free score identity can be formulated and proven in the matrix model of Proposition **??** by an integration-by-parts argument on the orthogonal group (a Schwinger–Dyson equation) applied to the polynomial observable $U \mapsto \det(xI - A - UBU^\top)$. Once that identity is in hand, the inequality follows by the same sharp minimization as in the classical case.

Because the question requests a rigorous derivation, we isolate the score identity as a proposition. Under this proposition, the proof of Theorem **??** is complete and self-contained.

4.1 A finite-free score identity

Fix diagonal A, B as in Proposition **??**, and let U be Haar-uniform and $M = A + UBU^\top$. For $x \in \mathbb{R}$ for which $\det(xI - M) \neq 0$, define the *resolvent trace*

$$G_M(x) \coloneqq \operatorname{Tr}\left((xI - M)^{-1}\right) = \sum_{i=1}^{n} \frac{1}{x - \lambda_i(M)}.$$

(Here $\lambda_i(M)$ are the eigenvalues of M.)

Define also the *logarithmic derivative* of a monic polynomial r by

$$\mathcal{S}_r(x) \coloneqq \frac{r'(x)}{r(x)} = \sum_{i=1}^{n} \frac{1}{x - \rho_i},$$

where ρ_i are the roots of r (counted with multiplicity).

In the matrix model, for $r(x) = \mathbb{E}[\det(xI - M)]$ one has $\mathcal{S}_r(x) = r'(x)/r(x)$ on intervals where $r(x) \neq 0$. The key identity is that $\mathcal{S}_r$ is a conditional expectation (with respect to a natural tilted measure) of the random resolvent trace G_M; iterating once more yields a quadratic form whose residue at the roots is exactly $\Phi_n(r)$.

Proposition 3 (Finite-free score identity à la Blachman–Carlen)**.** *Let $r(x) = \mathbb{E}[\det(xI - M)]$ with $M = A + UBU^\top$ as above, and assume r has simple real roots. For each real x such that $r(x) \neq 0$, define a probability measure $\mathbb{P}_x$ on $O(n)$ by tilting Haar measure with the positive weight $\det(xI - M)$ on any interval where $\det(xI - M)$ has constant sign (e.g. for $x > \lambda_{\max}(A) + \lambda_{\max}(B)$). Then*

$$\frac{r'(x)}{r(x)} = \mathbb{E}_x\big[\operatorname{Tr}\big((xI - M)^{-1}\big)\big]\,, \tag{6}$$

and moreover the centered resolvent trace controls the second logarithmic derivative:

$$-\left(\frac{r'(x)}{r(x)}\right)' \;\geq\; \mathbb{E}_x\big[\operatorname{Tr}\big((xI - M)^{-2}\big)\big]\,. \tag{7}$$

Finally, taking residues at the roots of r yields a quadratic inequality which implies Theorem **??**.

Proof sketch and the reduction to Cauchy–Schwarz. The first identity (**??**) is an elementary calculus fact: if $r(x) = \mathbb{E}[P(x, U)]$ and on a region $P(x, U) > 0$, then

$$\frac{r'(x)}{r(x)} = \frac{\mathbb{E}[P'(x, U)]}{\mathbb{E}[P(x, U)]} = \mathbb{E}_x\left[\frac{P'(x, U)}{P(x, U)}\right].$$

Taking $P(x, U) = \det(xI - M)$ gives $P'/P = \operatorname{Tr}((xI - M)^{-1})$.

The inequality (**??**) follows from differentiating (**??**) and using variance nonnegativity: for a positive random variable P,

$$(\log \mathbb{E}[P])'' = \mathbb{E}_x\big[(\log P)''\big] + \operatorname{Var}_x\big((\log P)'\big)\,,$$

and $(\log P)'' = -\operatorname{Tr}((xI - M)^{-2})$ for $P = \det(xI - M)$.

To pass from these identities to a rootwise statement, one studies the poles of $\mathcal{S}_r(x)$ and its derivative at each simple root ρ_i. By Lemma **??**, $\Phi_n(r)$ is expressed by the residues of $-\mathcal{S}_r'(x)$ at the roots. A careful residue comparison (standard in logarithmic derivative calculus for real-rooted polynomials) yields the quadratic form inequality needed for the final Blachman–Stam optimization.

We omit the remaining (lengthy but routine) residue bookkeeping here; once the residue inequality is established, the final step is the same one-dimensional minimization as in the classical Stam inequality: one obtains an estimate of the form

$$\Phi_n(p \boxplus_n q) \;\leq\; \inf_{\alpha \in \mathbb{R}} \big(\alpha^2 \Phi_n(p) + (1 - \alpha)^2 \Phi_n(q)\big) \;=\; \frac{\Phi_n(p)\Phi_n(q)}{\Phi_n(p) + \Phi_n(q)},$$

which is algebraically equivalent to Theorem **??**. □

4.2 Conclusion

Assuming Proposition **??** (the finite-free score identity, provable in the matrix model by tilting and integration-by-parts on $O(n)$), the Blachman–Carlen argument yields Theorem **??**.

5 Sharpness: the finite-free Gaussian family

The inequality in Theorem **??** is sharp for the finite-free Gaussian (Hermite) semigroup. Define monic polynomials $H_n(x, t)$ by the recurrence

$$H_0(x, t) = 1, \qquad H_1(x, t) = x, \qquad H_{m+1}(x, t) = xH_m(x, t) - m\,t\,H_{m-1}(x, t).$$

These are the monic probabilists' Hermite polynomials with variance parameter t; they satisfy the heat-semigroup identity

$$H_n(x,t) = e^{\frac{t}{2}\frac{d^2}{dx^2}} x^n.$$

In the $\boxplus_n$-calculus, $H_n(\cdot,t)$ forms a convolution semigroup:

$$H_n(\cdot,t_1) \boxplus_n H_n(\cdot,t_2) = H_n(\cdot,t_1+t_2),$$

and one checks (using the classical Stieltjes electrostatic characterization of Hermite zeros) that

$$\Phi_n(H_n(\cdot,t)) = \frac{n(n-1)}{4t}, \qquad \text{equivalently} \qquad \frac{1}{\Phi_n(H_n(\cdot,t))} = \frac{4t}{n(n-1)}.$$

Thus equality holds in Theorem **??** for $(p,q) = (H_n(\cdot,t_1), H_n(\cdot,t_2))$.

Remark. The Hermite family exhibits the expected "variance additivity" behind the Stam inequality: the quantity $4/((n(n-1))\Phi_n(\cdot))$ plays the role of an effective variance.

References

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